

Lian Aiwei

Phone number: +86 18983812155 | Email: 2023190505038@std.uestc.edu.cn | Address: UESTC, Chengdu, China

EDUCATION

University of Electronic Science and Technology of China (UESTC) Sep 2023-Jun 2027 (Expected)
B.Eng. in Electronic Information Engineering, Glasgow College

- **GPA:** 3.95/4.00; **Average:** 90.86/100; **Rank:** 6/234 (after five semesters)
- **IELTS:** 7.5 (Listening 8.0, Reading 8.0, Writing 7.5, Speaking 7.0)
- **Coursework:** Artificial Intelligence and Machine Learning, Dynamics and Control, Embedded Processors, Python Language Programming and Its Application, Probability Theory and Mathematical Statistics, Calculus.
- **Honors:** **National Scholarship (Year1, Year2)**; First-Class Scholarships from UESTC and Glasgow College (Year1, Year2)

Research Interests: Autonomous Robotics, Embodied Intelligence, Vision-Language Navigation

RESEARCH EXPERIENCE

Hong Kong University of Science and Technology (Guangzhou) Jul 2026 - Present
Research Intern, Intelligent Autonomous Driving Center

- Reproduced SysNav in simulation and conducted analysis of its semantic mapping, room-level topology, TARE-based exploration, and execution interfaces. Integrated camera, LiDAR, and Jetson on Go2 and complete the system integration on the physical platform.
- Analyzed the task-conditioned candidate scoring in ApexNav and the long-term semantic memory in DualMap. Currently developing a modular extension that retains SysNav's execution backbone, and adapting SCAN-Planner for local motion planning.

Tianfu Jiangxi Embodied Intelligence Perception Team Jan 2026 - Present
Localization Research Intern, UESTC, Chengdu

- Deployed and evaluated VINS-based methods, cuVSLAM, and OKVIS2-X on public and self-collected sequences; analyzed their visual-inertial estimation and optimization pipelines and constructed prior 3D Gaussian maps using Scaffold-GS.
- Adapted a lightweight GSCPR pipeline to establish 2D-3D correspondences through image matching, rendered-depth back-projection, and PnP/RANSAC. Integrated quality-gated prior-map reprojection residuals into the sliding-window optimization and marginalization of VINS-Mono, with migration to OKVIS2-X in progress.

PROJECT EXPERIENCE

SCOUT Mobile Robot Development Oct 2025 - Present
Core Team Member, UESTC, Chengdu

- Integrated a stereo camera, LSLiDAR-C32, and Jetson AGX Orin; completed URDF modeling and multi-sensor spatial and temporal calibration. Deployed Fast-LIVO2 for LiDAR-visual-inertial odometry and mapping and MM3DGS for 3D scene reconstruction.
- Implemented relocalization and navigation with Cartographer and Navigation2, and integrated EGO-Planner with Fast-LIO2 for navigation without a prior map. Deployed VLFM, StreamVLN, and a local Qwen model to investigate semantic decision interfaces and onboard inference latency.

RAICOM Robotics Developer Competition Apr 2025 - Aug 2025
Team Leader, National First Prize

- Led the development of an autonomous orchard mobile-manipulation system. Built a state-machine architecture and implemented STM32-Raspberry Pi communication with CRC validation and timeout handling, together with infrared-based local alignment.
- Trained and deployed a YOLOv5 fruit detector on an RGB-D camera through ONNX; recovered target positions in the robot frame and integrated detection, localization, grasping, and task-state transitions into a complete workflow.

Formula Student Electric China Dec 2023 - Oct 2024
Aerodynamics Team Member, National Third Prize

- Modeled front-wing and rear-wing aerodynamic components in CATIA and evaluated their performance through STAR-CCM+ CFD simulations; participated in material selection, component fabrication, assembly, and experimental validation, gaining hands-on experience across mechanical design and physical prototyping.

UAV Online Motion Planning Sep 2024 - Jun 2025
Research Training

- Studied differential flatness, RRT/RRT*, heuristic search, spline optimization, and MINCO trajectory parameterization; reproduced and analyzed Fast-Planner, EGO-Planner, and SVSDF in ROS, Gazebo, and RViz.

TECHNICAL SKILLS

- **Programming and Systems:** C/C++, Python, ROS 1/2, Linux
- **Robotics Platforms:** Jetson AGX Orin, STM32, Raspberry Pi, Gazebo, RViz
- **Engineering Tools:** CATIA, SolidWorks, STAR-CCM+, MATLAB